

PgBee FINAL REPORT

PgBee Business Growth & Expansion Analytics

Prepared By: Sidharth

Project Type: Business Analytics & Expansion Strategy

Organization: PgBee

Year: 2026

CERTIFICATE

This is to certify that the project titled "**PgBee Business Growth & Expansion Analytics**" is an original work conducted to analyze business performance, financial viability, forecasting, and expansion opportunities for PgBee.

ACKNOWLEDGEMENT

I would like to express my gratitude to all faculty members, mentors, hostel owners, students, and stakeholders who contributed to the successful completion of this project.

Special thanks to the PgBee beta participants from Thiruvananthapuram who provided valuable insights and validation of the business model.

ABSTRACT

PgBee is a student hostel discovery and booking platform developed to simplify accommodation search while helping hostel owners improve occupancy and visibility.

The project evaluates business performance using SQL analytics, R analytics, forecasting models, financial modeling, and business intelligence dashboards.

The beta launch successfully onboarded 38 hostels and generated 74 bookings, resulting in ₹36,926 revenue and ₹26,926 profit. Based on market analysis and forecasting, a phased expansion strategy is proposed across Kerala and later South India.

TABLE OF CONTENTS

1. Introduction
 2. Business Discovery
 3. Market Research
 4. Competitor Analysis
 5. KPI Framework
 6. SQL Analytics
 7. R Analytics
 8. Financial Modeling
 9. Dashboard Analysis
 10. Forecasting
 11. Expansion Strategy
 12. Recommendations
 13. Conclusion
 14. References
 15. Appendices
-

CHAPTER 1 – INTRODUCTION

Background

Student accommodation remains a major challenge for students relocating for education.

Most students rely on:

- Local brokers
- Social media groups
- Word-of-mouth recommendations

These methods are often inefficient and unreliable.

Problem Statement

Students face:

- Limited hostel information
- Lack of verified listings
- Time-consuming search processes

Hostel owners face:

- Low visibility
 - Vacant rooms
 - Difficulty reaching students
-

Proposed Solution

PgBee provides:

- Verified hostel listings
 - Digital discovery platform
 - Student-hostel matching
 - Booking support
-

Objectives

1. Analyze beta business performance.
 2. Evaluate financial viability.
 3. Forecast future growth.
 4. Identify expansion opportunities.
 5. Recommend growth strategies.
-

CHAPTER 2 – BUSINESS DISCOVERY

Business Model

PgBee acts as an intermediary platform.

Stakeholders

Students

Need:

- Affordable accommodation
- Verified information
- Easy booking

Hostel Owners

Need:

- Occupancy improvement
- Digital visibility
- Lead generation

PgBee

Generates revenue through:

Booking × Commission

Revenue Model

Commission Per Booking:

₹499

CHAPTER 3 – MARKET RESEARCH

Industry Overview

The student accommodation industry continues to expand due to:

- Rising college enrollments
 - Student migration
 - Urban education hubs
-

Target Market

Primary Market

Engineering students

Secondary Market

- Medical students
 - Arts & Science students
 - Professional course students
-

Kerala Market Opportunity

Major education hubs:

1. Thiruvananthapuram
 2. Ernakulam
 3. Kozhikode
 4. Thrissur
 5. Kollam
-

CHAPTER 4 – COMPETITOR ANALYSIS

Competitors

MagicBricks

Strength:

- Large property database

Weakness:

- Not hostel-focused
-

Housing.com

Strength:

- Strong brand

Weakness:

- Limited student specialization
-

Local Brokers

Strength:

- Local knowledge

Weakness:

- Lack of scalability
-

PgBee Advantage

Feature	PgBee
Hostel Focus	Yes
Student Focus	Yes
Verified Listings	Yes
Digital Platform	Yes

CHAPTER 5 – KPI FRAMEWORK

Growth KPIs

- Total Hostels
 - Total Bookings
 - Revenue
-

Operational KPIs

- Conversion Rate
 - Booking Success Rate
 - Hostel Utilization
-

Financial KPIs

- Profit
 - Margin
 - CAC
 - LTV
-

CHAPTER 6 – SQL ANALYTICS

Database Design

Tables:

students

hostels

hostel_owners

bookings

Beta Results

Total Hostels

```
SELECT COUNT(*) FROM hostels;
```

Result:

38

Total Bookings

```
SELECT COUNT(*) FROM bookings;
```

Result:

74

Revenue

Formula:

$74 \times ₹499$

Result:

₹36,926

SQL Findings

- Stable booking activity
 - Positive demand validation
 - Revenue generation achieved
-

CHAPTER 7 – R ANALYTICS

Data Cleaning

Performed:

- Missing value checks
- Duplicate removal
- Data validation

Outputs:

students_clean.csv
hostels_clean.csv
bookings_clean.csv

Descriptive Analytics

Calculated:

- Student counts
 - Hostel counts
 - Booking counts
-

Customer Segmentation

Technique:

K-Means Clustering

Variables:

- Rent Preference
 - Booking Value
-

Forecasting

Technique:

ARIMA Forecasting

Purpose:

Predict future booking demand.

CHAPTER 8 – FINANCIAL MODELING

Revenue Analysis

Formula:

Revenue = Bookings × ₹499

Current Revenue:

₹36,926

Cost Structure

Category	Amount
Marketing	₹5,000
Hosting	₹2,000
Admin	₹3,000

Profitability

Profit:
₹26,926
Margin:
72.92%

Unit Economics

KPI	Value
CAC	₹67.57
LTV	₹499
LTV/CAC	7.39

Break-Even Analysis

Break-Even Bookings: 21
Current Bookings: 74
Status: Above Break-Even

CHAPTER 9 – DASHBOARD ANALYSIS

Power BI Dashboard

KPI Cards

- Hostels
 - Bookings
 - Revenue
 - Profit
-

Funnel Analysis

Visitors → Logins → Bookings
283 → 153 → 74

Waterfall Analysis

Revenue contribution visualization.

Revenue Allocation

Distribution of:

- Profit
 - Marketing
 - Hosting
 - Administration
-

CHAPTER 10 – FORECASTING

Forecast Objective

Estimate future booking demand.

Forecast Results

Phase 1

Bookings:

150

Phase 2

Bookings:

250

Phase 3

Bookings:

180

Forecast Insight

Significant growth potential exists through geographic expansion.

CHAPTER 11 – EXPANSION STRATEGY

Kerala Scoring Model

Top Districts

Rank	District
1	Thiruvananthapuram
2	Ernakulam
3	Kozhikode

Expansion Roadmap

Phase 1

Thiruvananthapuram

Target:

- 50 Hostels
 - 150 Bookings
-

Phase 2

Ernakulam

Target:

- 75 Hostels
 - 250 Bookings
-

Phase 3

Kozhikode

Target:

- 50 Hostels
 - 180 Bookings
-

India Expansion

Priority States:

1. Kerala
 2. Karnataka
 3. Tamil Nadu
 4. Telangana
 5. Andhra Pradesh
-

CHAPTER 12 – RECOMMENDATIONS

Short-Term

- Increase hostel inventory
 - Improve conversion rates
 - Expand marketing campaigns
-

Medium-Term

- Launch Ernakulam operations
 - College partnerships
 - Referral programs
-

Long-Term

- South India expansion
 - Mobile application
 - Marketplace scaling
-

CHAPTER 13 – CONCLUSION

Key Results

KPI	Value
Hostels	38
Bookings	74
Revenue	₹36,926
Profit	₹26,926

Final Conclusion

The PgBee beta successfully validated the platform's business model and demonstrated strong potential for growth.

Financial analysis confirms profitability, forecasting indicates scalability, and expansion analysis identifies clear opportunities within Kerala.

Final Recommendation

Phase 1 → Strengthen Thiruvananthapuram

Phase 2 → Launch Ernakulam

Phase 3 → Launch Kozhikode

Future Goal:

Build South India's leading
student accommodation platform.